

Vitals — Run of Show

Nine slides, spoken — 5:20 of words at 145wpm, which lands near five minutes at a real presenting pace. Written to be read off a screen while standing up, so the sentences are short and the clauses don't nest. Anything in the script you can't say out loud in one breath, cut.

RUNTIME **5:20** 9 SLIDES 60-SECOND CUT INCLUDED DEMO IS LIVE, NOT RECORDED
BUILT FOR A **5-MINUTE** SLOT

BEFORE YOU RECORD

- **Run the demo once, cold, before the take.** The rejection on slide 5 has to be the program actually saying no, not a screenshot. If it's a screenshot, say so.
- **The first fifteen seconds decide it.** Judges have hundreds of these. Open on the patient, not on the problem statement.
- **Slow down on the three hashes.** That's the moment the claim becomes checkable. Let it sit.
- **Timings are measured, not estimated.** Each clock is that beat's actual word count at 145 words a minute. If you naturally run faster, the whole thing shortens proportionally — don't try to hit the marks.
- Never say "certificate", "badge", or "credentials on the blockchain" — every judge has a reflex against all three.

01 Cold open

0:00 – 0:18 · 18s

ON SCREEN Title slide. Or better: the game already running, monitor visible, vitals moving.

A nineteen-year-old walks into an emergency room. Ten minutes ago she ate something with shrimp in it. Her throat is closing.

You have about four minutes. The simulation will not wait for you.

That's Vitals. **Anyone can play it. Nobody can fake the replay.**

DELIVERY Do not introduce yourself first. Do not say "hi, we're building". Start inside the case — the judge should be in the room before they know what they're watching.

02 The problem

0:18 – 0:49 · 31s

ON SCREEN Slide 02 — 4,350+ schools · ~23 countries · 130–147 countries applying

Every medical student runs hundreds of encounters like that. Every one is scored. Every score is thrown away.

Since 2024, to practise in the US, your school needs a WFME-recognised accreditor. About twenty-three countries have one. There are over four thousand medical schools on earth.

So most graduates alive cannot prove what they can do — not because nobody measured them, but because nobody can check it. **The evidence exists. The trust layer doesn't.**

DELIVERY Land "twenty-three" and "four thousand" against each other. That gap is the whole slide — don't rush past it to get to the technology.

03 It already runs

0:49 – 1:33 · 44s

ON SCREEN Slide 03 — three tapes, three results, then the three hashes

This isn't a mockup. Three people played that case.

The first gives adrenaline, oxygen, lays her flat, keeps her for observation. She lives.

The second gives the adrenaline, then stands her up. In anaphylaxis the ventricle is empty — standing someone up can kill them. She survives, but the harm is on the record.

Look at the hashes. **Same outcome. Completely different leaf.** We didn't record whether she lived. We recorded how you got there.

The third gives antihistamines and waits.

— PAUSE —

She dies. Nobody scripted that. There's no losing branch in a story tree — the physiology killed her, the way it would have.

DELIVERY This is the longest beat and it earns it. Pause after "stands her up" — a clinician in the room will react, and that reaction sells the fidelity better than you can.

04 The mechanism

1:33 – 2:15 · 42s

ON SCREEN Slide 04 — commit · play · reduce · anchor, then the leaf formula

Every run reduces to one hash.

You commit before you play, so the chain knows how many attempts you **started**. Grinding for a lucky run is visible.

Then you play a deterministic automaton. The tape is what you did and when. We reduce it to discrete facts — beats, harm, outcome — and anchor one leaf. Trajectories are floating point, so we don't anchor those. **The trajectory is simulated. The outcome is proven.**

And there's no language model anywhere in that path. Chess engines have verified replays for decades. We point it at a patient who's dying on a clock.

DELIVERY "No language model in that path" is the line that separates you from every other AI-plus-crypto submission this round. Say it flat, like a spec, not like a boast.

05 The refusal

2:15 – 2:46 · 31s

ON SCREEN Slide 05 — live terminal preferred: claim Proficient → REJECTED, claim Competent → GRANTED

Here's my favourite part. Watch the program refuse me.

I claim Proficient. It recomputes the level with the same arithmetic the game uses and says no. **You're Competent. Three distinct cases.**

Proficient needs five. That threshold isn't a demo constant — it's been in the shipping competency model for a year, and it's now refusing me on chain.

I claim Competent. Granted. No issuer key, no oracle, no authority anyone can lean on or subpoena.

DELIVERY Run it live if you possibly can. A protocol refusing its own author is the single most persuasive thing in this deck, and a screenshot of it is worth about a tenth as much.

06 Onchain architecture

2:46 – 3:11 · 25s

ON SCREEN Slide 06 — four primitives table, then volume / latency / onboarding

Four primitives, none invented by us. Registry, compressed anchors, non-transferable progression, Attestation Service.

Why Solana: a cohort year worldwide is hundreds of millions of runs — a hundred and ten dollars per million, compressed. Anchoring only the exams would kill the idea.

And the player never sees a seed phrase. A nineteen-year-old learning medicine shouldn't have to learn key management first.

DELIVERY Fastest beat in the deck. If you're running long, this is where you cut — trim to the cost number and the seed-phrase line.

ON SCREEN Slide 07 — stake / verify / dispute loop, sinks-and-sources, never-tokenized list

Almost everything here is already trustless. But one thing isn't: somebody has to replay the tape and attest the outcome — and if that's one party, I've rebuilt the database I'm escaping.

So \$VITAL bonds an open verifier set. Operators stake, anyone can challenge by re-running the tape, the wrong one gets slashed.

Optimistic verification usually fails because most disputes are fuzzy. This one isn't. **Replay is exact — two honest verifiers cannot disagree.** The determinism that makes the credential mean something is what makes the token enforceable.

And the list I care about more: credentials and badges are never tradeable, nothing gates access to play, nobody buys a better replay. A liquid market in "Expert in Cardiology" would destroy this faster than any attack.

No token ships in this hackathon. A verification market only needs to exist once verification is worth attacking.

DELIVERY The last two paragraphs are the ones that land. Every judge has sat through a hundred token slides today; almost none of them heard a team say what they refuse to tokenize, or that they're not launching yet.

08 Precedent, and who we are

4:10 – 4:50 · 39s

ON SCREEN Slide 08 — CrowdBrain quote, then the head-start card

The Grand Champion of Frontier this year was CrowdBrain: train users in simulation, qualify them through QA, route the best operators to real robots. **That's this, for robot operators instead of clinicians.** Out of two thousand eight hundred submissions.

And we didn't start four weeks ago. The simulator underneath is in production with real students — four hundred and twenty-four scenarios, deterministic scoring, a hash-chained audit log, mapped to the AAMC framework.

Most teams spent this hackathon building the thing that produces the signal. **We already had it. We spent it making the signal checkable.**

DELIVERY Say it as a compliment to them, not a claim about you. Then hand off — **this is where a second team member should speak**, and where names and roles go. Colosseum's own data says winning teams average three or more people across technical and non-technical roles, and a solo voice for five straight minutes reads as a solo project.

ON SCREEN Slide 09 — working now / four weeks / cut-on-purpose

Two halves are built. The replay produces leaves. The program refuses bad claims. One week joins them.

What we want from this room: the accreditors and residency programmes who would actually rely on a record like this. We can build the protocol. We can't grant it standing.

— PAUSE —

You play a patient who's dying, the chain keeps a replay of what you did, and anyone can re-derive it without asking us.

DELIVERY The ask goes before the pause, never after. End on the one-sentence version and stop — no thank-you, no roadmap. The silence does more than another clause will.

The 60-second cut

For the pitch video's opening, or any time you get cut short. Slides 1, 3, 5, 9 only. If a judge understands nothing else, they should come away with these four things: the stakes are real, it runs, the chain refuses lies, and anyone can check it.

A nineteen-year-old, ten minutes after eating something with shrimp in it. Her throat is closing. You have four minutes and the simulation won't wait.

Three people played that case. One saved her. One saved her but stood her up first — same outcome, different hash, because the harm is on the record. One gave antihistamines and she died. Nobody scripted that.

Every run reduces to one hash on Solana. No language model anywhere in the proof path.

Then watch this: I claim Proficient, and the program recomputes it and refuses me. No issuer key. No oracle. The level on chain is the one it computed, not the one I claimed.

You play a patient who's dying, the chain keeps a replay of what you did, and anyone can re-derive it without asking us.

Questions you will get, and the answers you should already have

WHY DOES THIS NEED A BLOCKCHAIN?

Because the verifier can't be the same party as the scorer, and the person relying on the record is in another country. Remove the chain and you're back to one company's database.

HOW DO YOU KNOW THE STUDENT TOOK THE EXAM?

We don't, and we say so. We prove the scoring was faithful to the tape. Identity binding is the issuing institution's job, and the credential names them so you can price that trust yourself.

ISN'T THIS JUST AN NFT CERTIFICATE?

Show the refusal. A certificate can't reject the person holding it.

WHAT ABOUT THE LLM PARTS?

The patient's dialogue in other modes. Never the evidence. In story mode it's a keyword graph, so even that is deterministic.

IS THE TOKEN A SECURITY?

It bonds computation and punishes lying about computation. It confers no qualification and no access to care. That line is deliberate and it's written down.

WHO'S ACTUALLY PAYING?

Institutions on ordinary invoices and sponsors funding outcome-linked scholarships. Players pay nothing and hold nothing.